WORKING THESIS DRAFT

Design and Robustness Analysis of a Low-Power Hybrid FS-GDI/CMOS Extended Hamming (12,7) Encoder

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Abstract

This working thesis draft evaluates CMOS, GDI, FS-GDI, and hybrid FS-GDI/CMOS implementations of Hamming (11,7) and extended Hamming (12,7). The baseline paper is independently audited, SEC-DED logic is exhaustively verified, and 121 LTspice experiments are executed with a predictive 65-nm PTM model. Of these, 118 measurement sets pass automated validation and three low-load cases are rejected for excessive overshoot. Hybrid-B provides the lowest measured D1-to-P0 delay at the nominal condition, whereas FS-GDI retains the lowest simulated power and PDP.

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Abbreviations

Abbreviation	Meaning
CMOS	Complementary metal-oxide-semiconductor
GDI	Gate diffusion input
FS-GDI	Full-swing gate diffusion input
PDP	Power-delay product
PTM	Predictive technology model
SEC-DED	Single-error correction, double-error detection

Chapter 1 - Introduction

Low-power error-control hardware is dominated by XOR networks. Device count alone does not establish useful operation because pass-transistor degradation, output load, restoration overhead, and cascade depth can change the power-delay trade-off. The objective is to reproduce the baseline Hamming network and test whether selective CMOS restoration produces a defensible extended-Hamming implementation.

Research objectives

The work audits the published equations and percentages, verifies extended Hamming logic, compares cells and complete encoders under common conditions, measures energy and signal swing, and maps controlled voltage, temperature, frequency, and load variation.

Chapter 2 - Literature Survey

Hamming established the coding theory; Morgenshtein and co-authors established GDI and later FS-GDI. Recent GDI work emphasizes adders, approximate arithmetic, and emerging devices, while recent ECC work addresses SEC-DED and stronger adjacent-error codes. Extended Hamming and GDI are therefore not new by themselves.

Research gap and motivation

The closest published Hamming study does not provide extended parity, machine-readable netlists, or a complete robustness map. The gap addressed here is a traceable one-model comparison with explicit P0 timing, common activity, selective restoration, and retained negative results.

Chapter 3 - Theoretical Background

Parity bits occupy positions 1, 2, 4, and 8; data occupy positions 3, 5, 6, 7, 9, 10, and 11; P0 is appended at position 12.

$$P1 = D1 \text{ XOR } D2 \text{ XOR } D4 \text{ XOR } D5 \text{ XOR } D7 \quad (3.1)$$

$$P2 = D1 \text{ XOR } D3 \text{ XOR } D4 \text{ XOR } D6 \text{ XOR } D7 \quad (3.2)$$

$$P4 = D2 \text{ XOR } D3 \text{ XOR } D4; \; P8 = D5 \text{ XOR } D6 \text{ XOR } D7 \quad (3.3)$$

$$P0 = D1 \text{ XOR } \ldots \text{ XOR } D7 \text{ XOR } P1 \text{ XOR } P2 \text{ XOR } P4 \text{ XOR } P8 \quad (3.4)$$

SEC-DED requires an appropriate decoder. The encoder alone does not correct errors.

Chapter 4 - Baseline Reproduction

The Hamming (11,7) network uses twelve XOR2 cells. The resulting CMOS and FS-GDI counts of 144 and 72 match the baseline paper's structural counts. Numerical matching is not expected because the original model deck, load, input pattern, and exact measurement definitions were not released.

Architecture	Power (uW)	Delay (ps)	PDP (fJ)	Swing (V)	Transistors	Status
CMOS	12.0805	164.049	1.9818	1.2077	144	Pass
GDI	0.7756	16089.790	12.4793	1.2043	48	Fail
FSGDI	0.5029	139.363	0.0701	1.2053	72	Pass

Chapter 5 - Extended Hamming (12,7)

The extended architecture adds a ten-XOR overall-parity path to the twelve-XOR Hamming network. The same Boolean graph is retained across logic styles to preserve comparison fairness.

Chapter 6 - Proposed Hybrid FS-GDI/CMOS Design

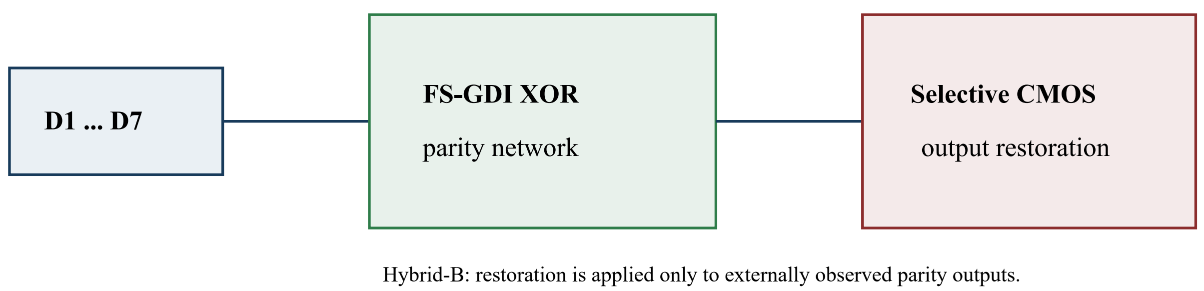

Figure 6.1. Original block diagram of Hybrid-B.

Hybrid-A restores every XOR output. Hybrid-B restores only the externally observed parity outputs. Hybrid-B contains 152 transistors, compared with 264 for CMOS, 132 for FS-GDI, and 220 for Hybrid-A.

Chapter 7 - LTspice Simulation Methodology

Parameter	Value
Simulator	LTspice 26.0.2
Model	PTM 65-nm bulk CMOS beta
Nominal	1.2 V, 27 C, 125 MHz, 10 fF
Sizing	NMOS 120/65 nm; PMOS 240/65 nm
Delay	50%-to-50%; D1-to-P0 for EH127
Power	AVG(-V(VDD)*I(VSUP))
Sweeps	VDD, temperature, load, and frequency independently

Chapter 8 - Results and Robustness Analysis

Exhaustive verification covered 128 data words, 1536 single-bit cases, and 8448 double-bit cases. All passed.

Architecture	Power (uW)	Delay (ps)	PDP (fJ)	Swing (V)	Transistors	Status
CMOS	27.0237	344.947	9.3217	1.2064	264	Pass
FSGDI	2.5502	429.062	1.0942	1.2135	132	Pass
HYBRID_A	25.8367	278.783	7.2029	1.2039	220	Pass
HYBRID_B	11.2029	215.846	2.4181	1.2042	152	Pass

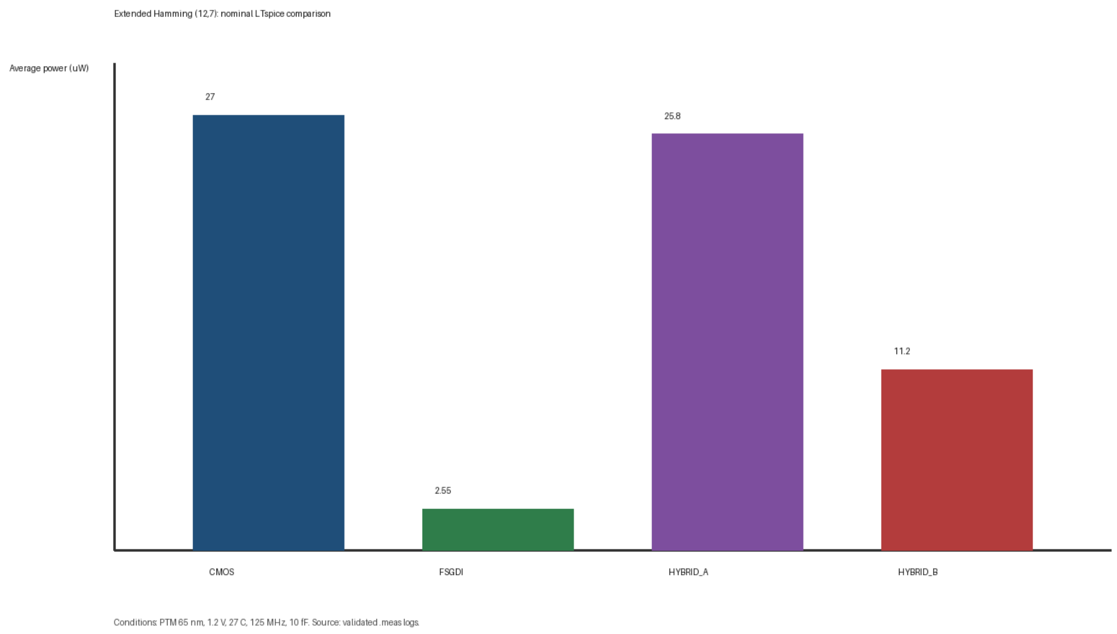

Figure 8.1. Nominal average power.

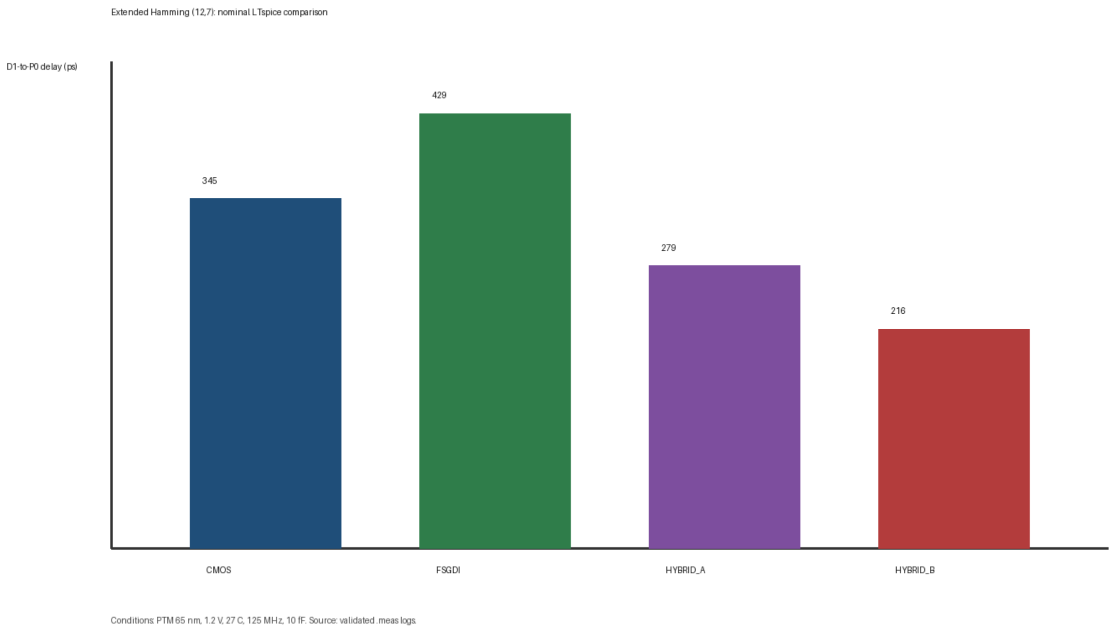

Figure 8.2. Nominal D1-to-P0 delay.

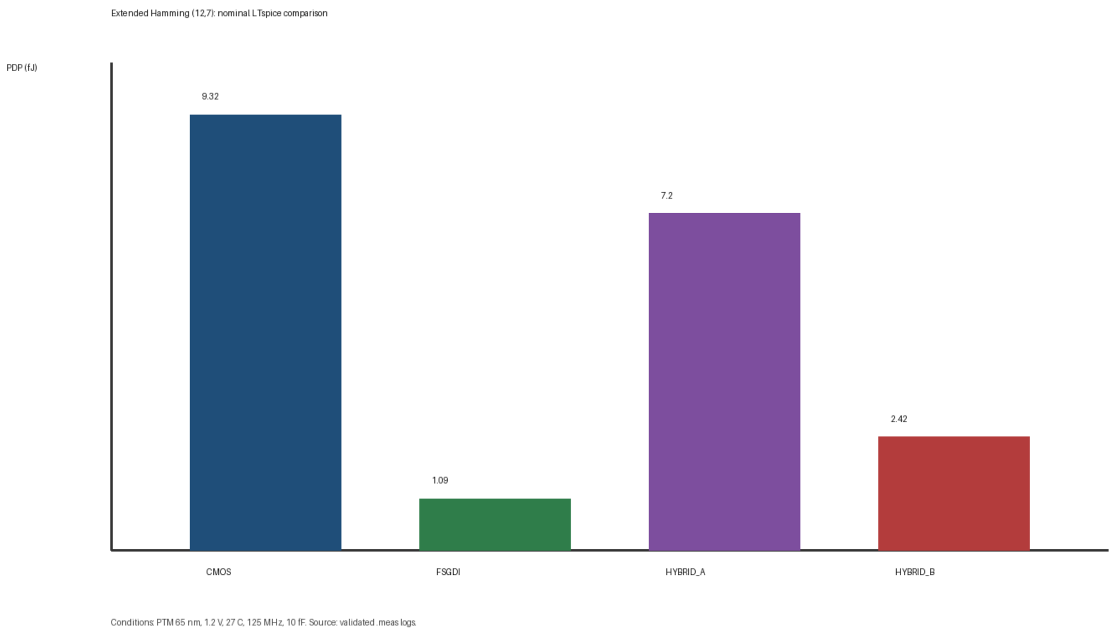

Figure 8.3. Nominal PDP.

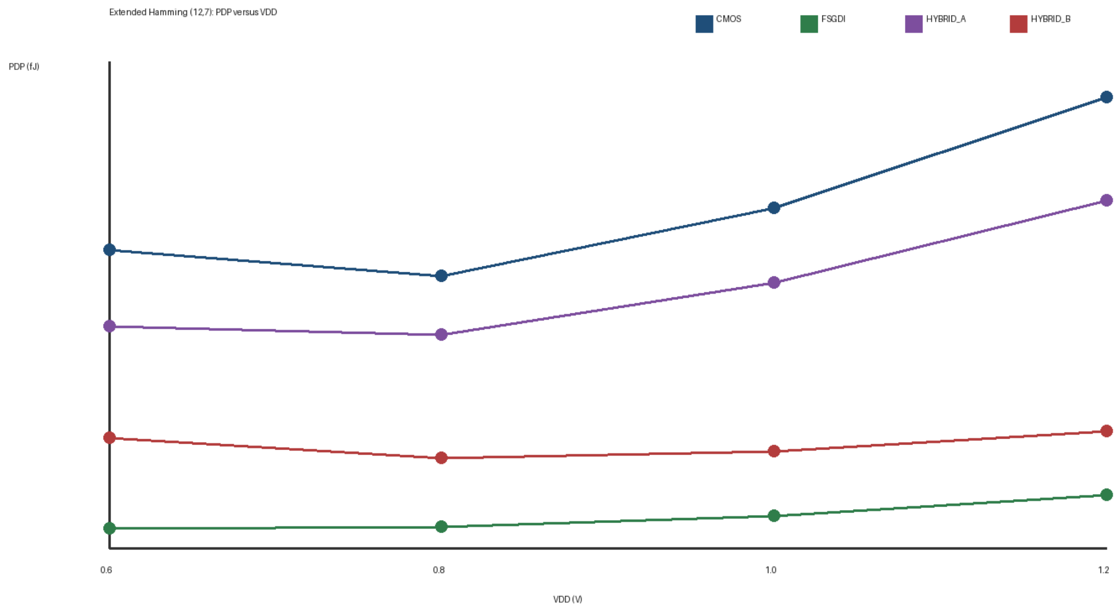

Figure 8.4. PDP versus supply voltage.

Chapter 9 - Discussion

At nominal conditions, Hybrid-B uses 11.203 uW and has a D1-to-P0 delay of 215.846 ps. Relative to CMOS, this is 58.54% lower power and 37.43% lower delay. FS-GDI uses only 2.550 uW but has a longer P0 path. Restoration therefore buys speed and external drive at an energy cost.

Three 1 fF simulations are rejected because overshoot exceeds the configured supply-bound tolerance. All retained architectures tested in the 10 fF voltage sweep satisfy the study criteria at 0.6 V; lower voltages were not tested, so no boundary below 0.6 V is claimed.

Chapter 10 - Conclusions and Future Work

Hybrid-B is supported as a speed-oriented selective-restoration candidate, not as a universal energy optimum. FS-GDI remains preferable when simulated energy and PDP dominate. Future work requires supervisor review of the reconstructed cell, an independent foundry-PDK or second-simulator cross-check, post-layout parasitics, and a lower-voltage boundary search.

References

[1] R. W. Hamming, 'Error Detecting and Error Correcting Codes,' Bell System Technical Journal, vol. 29, no. 2, pp. 147-160, 1950.

[2] A. Morgenshtein, A. Fish, and I. A. Wagner, 'Gate-Diffusion Input (GDI): A Power-Efficient Method for Digital Combinatorial Circuits,' IEEE TVLSI, 2002.

[3] A. Morgenshtein et al., 'Full-Swing Gate Diffusion Input Logic - Case-Study of Low-Power CLA Adder Design,' Integration, 2014.

[4] M. A. M. El-Bendary and O. El-Badry, 'FS-GDI Based Area Efficient Hamming (11,7) Encoding,' International Journal of Electronics, 2024.

Appendix A - Reproducibility

The repository preserves 121 circuit files, their LTspice logs, the PTM model, experiment manifest, Python processing scripts, Verilog reference models, processed datasets, and figures.

Appendix B - Status

This document is a WORKING THESIS DRAFT. University front matter, institutional metadata, supervisor approval, and any required disclosure statements remain pending.